EE4091: Photonics - Midterm Exam Duration: 60 minutes Solve any 5 questions

(1) A TM-polarized light wave is incident from a thick glass slab of refractive index 1.8 to air, with angle of incidence 30°. Determine the fraction of light power transmitted to air. What should the incidence angle be such that the incident power is totally reflected back in the glass slab?

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(2) The visible light complex refractive index curve of GaAs is shown in the figure. At red color energy (1.8 eV), find (a) the value of the complex refractive index, (b) the speed of light and (c) the attenuation coefficient. How far would a red light beam travel inside a GaAs material before it reduces to 1% of its original power density value?

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(3) Two in-phase spherical white light sources, 100 μm apart are projected on a screen 5 meters away. Calculate the separation between the first blue (470 nm) and red (680 nm) bright spots on the screen.

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(4) An unpolarized light beam of intensity 800 W/m^2, passes through a sequence of three polarizers. Measured from the vertical, the transmission axes of the polarizers are at angles of 20°, 50°, and 140°, respectively. What is the intensity of the light when it emerges from each polarizer?

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(5) A shaving mirror has a concave surface (R=45 cm). How far from the mirror should one's face be for the image to be 35 cm from the real face?

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(6) In the photoelectric effect experiment, a 0.5-W beam of blue light (450 nm) is incident on a sodium metal plate (W=2.4 eV) causing electrons to be ejected. Calculate the kinetic energy of the ejected electrons and the reading on the ammeter. Assume one photon ejects one electron

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